



Sri Lanka Institute of Information Technology

Web-Based Library Management System

Part A - Requirement Gathering

Year 2 - Semester 1 | June 2026

IT2140 - Database Design and Development

MLB-B11G2-10

Group Members

IT25101420	Rayyan M.R.M
IT25102212	Vithanage C.B
IT25103166	Lakshitha P.M.M
IT25100467	Wijewickrama S.N
IT25102574	Wijegunawardana P.K.D.S.U
IT25100041	Amukotuwa A.G.S.I

1. Requirement Gathering Process

This report documents the requirement-gathering work completed for the proposed Web-Based Library Management System. The group used the Project Discussion Bot to collect stakeholder perspectives from the Administrator, Librarian, Library Member, and Finance Officer. The discussions were used to identify supported functional requirements, non-functional requirements, stakeholder roles, and the major functions summarized below.

The terminology and function organization in this report have been aligned with the approved project proposal. The original stakeholder discussion screenshots are retained unchanged in Appendix A as the evidence collected during Lab 01.

2. System Purpose

The Web-Based Library Management System is an online platform developed to manage library operations efficiently. The system allows librarians to manage books, members, borrowing activities, returns, fines, and reports through a centralized web application. Library members can also search books online, check availability, and view their borrowing records.

The system aims to reduce manual paperwork, improve record management, and provide better services for both library staff and members. The main objectives of the system are to:

- Manage books and members efficiently.
- Support online book searching and availability checking.
- Record book borrowing and return activities accurately.
- Calculate fines for overdue books automatically.
- Generate reports for library management.

3. Users and Stakeholders

The following stakeholders and responsibilities are aligned with the approved project proposal.

Stakeholder	Role in the proposed system
Library Member	Searches books, borrows books, checks fines, and views borrowing records.
Librarian	Manages books, members, borrowing activities, and returns.
Administrator	Manages users, reports, fines, and system settings.
Finance Officer	Views and manages fine payments and financial reports.
Academic Staff Member	A specialized type of Library Member who can additionally reserve unavailable books and manage waiting-list renewals.

Stakeholder	Role in the proposed system
Guest Visitor	Browses the online catalogue, searches books, and views ratings/reviews without logging in.

4. Functional Requirements

Based on the stakeholder discussions and the approved proposal, the system should provide the following functions.

1. User Management: allow members, librarians, and administrators to create accounts, log in, manage profiles, manage accounts, and assign user roles.
2. Book Catalog and Inventory Management: add, update, search, and delete book records; manage book copies, categories, and availability.
3. Borrowing and Return Management: issue books to registered members, record borrowing details, record returns, update book availability, renew borrowed books, and track due dates.
4. Fine Management: calculate overdue fines, record payments, view outstanding fines, and update fine status.
5. Reservation Management: allow members to reserve books, view reservation status, cancel reservations, and support waiting-list management.
6. Online Book Search: allow members to search books by title, author, or category and check availability.
7. Reports and Analytics Management: generate reports on borrowed books, overdue books, fines, reservations, users, and inventory.
8. Book Ratings and Reviews Management: allow members to rate and review previously borrowed books, edit or delete their own reviews, and allow administrators to remove inappropriate reviews.
9. Member Feedback Management: allow members to submit, view, and edit feedback about library services, and allow administrators to review, respond to, and manage feedback status.
10. Fine Appeal and Waiver Management: allow members to submit a fine appeal with a reason and view its status, and allow administrators to approve or reject the appeal, reduce or waive the fine, and record the decision and comments.
11. Fine Payment and Receipt Management: allow members to make full or partial fine payments using a selected payment method, view the remaining balance and payment history, generate and view/print a unique receipt, and automatically update the fine status to Paid or Partially Paid.

5. Non-Functional Requirements

The proposed system should meet the following non-functional requirements.

Requirement	Description
Performance	The system should respond quickly and load pages within a few seconds.
Security	User passwords should be encrypted and access should be role-based.
Usability	The interface should be simple, clear, and easy to use.
Scalability	The system should support future growth and additional features.
Maintainability	The system should be easy to update and maintain.

6. Constraints and Assumptions

The following constraints and assumptions apply to the proposed Web-Based Library Management System.

Constraints

Constraint	Description
Web-Based Platform	Accessible only through a web browser.
Internet Dependency	Requires an active internet connection.
Data Accuracy	Depends on correct data entered by staff.
Browser Compatibility	Requires modern web browsers.

Assumptions

- Users have basic knowledge of web applications.
- Library staff enter accurate book and member information.
- Members use valid contact information for notifications.

7. Major Functions

The following ten major functions use the naming and organization approved in the project proposal.

7.1 User Management

Users	Supported features	Expected outcome
Members, Librarians, Administrators	Register users, log in, update profiles, manage accounts, and assign user roles.	Secure access and efficient user-account management.

7.2 Fine Management

Users	Supported features	Expected outcome
Librarians, Administrators	Calculate fines, view outstanding fines, record payments, and update fine status.	Accurate fine calculation and efficient payment tracking.

7.3 Reservation Management

Users	Supported features	Expected outcome
Librarians, Members	Reserve books, view reservation status, cancel reservations, and manage waiting lists.	Organized reservations and improved book availability.

7.4 Book Catalog and Inventory Management

Users	Supported features	Expected outcome
Librarians, Administrators	Add, update, search, and delete book records; manage book copies, categories, and availability.	Accurate, organized, and up-to-date book inventory.

7.5 Borrowing and Return Management

Users	Supported features	Expected outcome
Members, Librarians	Issue books, return books, renew borrowed books, and track due dates.	Efficient borrowing process and accurate book-availability tracking.

7.6 Reports and Analytics Management

Users	Supported features	Expected outcome
Administrators, Librarians	Generate reports on borrowed books, overdue books, fines, reservations, users, and inventory.	Better decision-making through accurate reports and system insights.

7.7 Book Ratings and Reviews Management

Users	Supported features	Expected outcome
Members, Administrators	Rate and review borrowed books, edit or delete own reviews, and remove inappropriate reviews.	Reliable book ratings and a moderated review system for all users.

7.8 Member Feedback Management

Users	Supported features	Expected outcome
Members, Administrators	Submit, view, and edit feedback; review, respond to, and manage feedback status.	Improved library services through tracked and actionable member feedback.

7.9 Fine Appeal and Waiver Management

Users	Supported features	Expected outcome
Members, Finance Officer, Administrators	Submit a fine appeal with a reason, view appeal status, approve or reject the appeal, reduce or fully waive the fine, and record the administrator's decision and comments.	Fair and transparent handling of disputed fines with a documented decision trail.

7.10 Fine Payment and Receipt Management

Users	Supported features	Expected outcome
Members, Finance Officer	Record full or partial payments, select a payment method, calculate the remaining balance, generate a unique receipt number, view and print receipts, view payment history, and automatically mark the fine as Paid when the balance reaches zero.	Accurate, transparent, and fully traceable fine payment processing.

8. Conclusion

The requirement-gathering process helped the group understand the needs of the identified stakeholders. The collected requirements have been organized into proposal-aligned stakeholder roles, functional requirements, non-functional requirements, and major functions for the Web-Based Library Management System.

These supported requirements will guide the design, development, testing, and implementation activities in the next phases of the project. The proposed system is intended to reduce manual work, improve record accuracy, and provide more efficient services for library users.

Stakeholder Discussion Evidence

Figure 1: Discussion with Library Administrator

What are the main functions and features you expect from a Web-Based Library Management System?



Project Discussion Bot - Year2

Mr. Nuwan Perera – Library Director (Library Administrator)

As the Library Administrator, I expect the Web-Based Library Management System to support all major library operations in one place.

The main functions I expect are:

Member Management

The system should allow staff to register new members, update member details, manage memberships, and view borrowing history.

Book Management

Staff should be able to add new books, edit book information, organise books by category, author, and ISBN, and manage the number of available copies.

Book Search and Catalogue

Users should be able to search for books easily using title, author, ISBN, category, or keywords. The system should show real-time availability.

Book Issuing and Returning

The system should record book loans, due dates, returns, and automatically update the availability status of books.

Book Reservation

Members should be able to reserve unavailable books and receive notifications when they become available.

Book Renewal

Members should be able to request extensions for borrowed books according to library rules.

Fine Management

The system should automatically calculate overdue fines, record payments, and maintain fine histories.

Lost and Damaged Book Management

Staff should be able to record lost or damaged books, calculate compensation charges, and maintain related records.

User and Role Management

Different access levels should be available for Administrators, Librarians, and Library Members.

Reports and Analytics

The system should generate reports on book inventory, borrowing activities, overdue books, member registrations, reservations, fines, and library performance.

Notifications and Alerts

The system should send reminders for due dates, overdue books, reserved books, and important library announcements.

Security and Backup

The system should provide secure login, access control, activity logs, and regular database backups.

Overall, I expect the new system to reduce paperwork, improve accuracy, save staff time, provide better service to members, and allow us to manage the library more efficiently.

Figure 2: Administrator Dashboard Requirements

What information should appear on the Administrator Dashboard, and what reports should the system generate?



Project Discussion Bot - Year2

Mr. Nuwan Perera – Library Director (Library Administrator)

For the Administrator Dashboard, I would like to see the most important information as soon as I log in.

The dashboard should show the total number of books in the library, total registered members, books currently borrowed, books returned today, and overdue books. It should also display recent library activities such as new member registrations, new books added, and recent borrowing transactions.

I would like to see alerts and notifications for overdue books, unpaid fines, low book stock, and pending reservations. A simple chart showing monthly borrowing trends would also help me understand library usage.

I also expect a reservation feature. Members should be able to reserve books that are currently unavailable.

The system should calculate fines automatically for late returns and keep records of all payments.

We need search facilities so users can quickly find books by title, author, category, or keyword.

Reports are very important. I would like reports on book inventory, borrowed books, overdue books, member activities, and fine collections.

The system should have different user roles such as Administrator, Librarian, and Library Member, with different access permissions.

Finally, I expect the system to be web-based, easy to use, secure, and accessible from any location with an internet connection.

These are the main functions I would expect before approving a Library Management System for our organisation.

For reports, the system should generate:

- Book Inventory Report
- Borrowed Books Report
- Returned Books Report
- Overdue Books Report
- Reserved Books Report
- Member Registration Report
- Member Activity Report
- Fine Collection Report
- Most Borrowed Books Report
- Lost or Damaged Books Report
- Daily, Monthly, and Yearly Library Activity Reports

These reports should allow filtering by date, category, member, and book title. I would also like the option to print or download reports for management meetings and record keeping.

These reports will help us monitor library performance, make purchasing decisions, and ensure better operations run more efficiently.

Figure 3: Librarian Requirement Gathering

Explain the complete process of registering members, managing books, issuing books, returning books, and handling lost or damaged books.



Project Discussion Bot - Year2

Mr. Nuwan Perera – Library Director (Library Administrator)

From our library's point of view, the process would work like this:

Member Registration

When a new person wants to join the library, the librarian collects their details such as name, address, phone number, email, and identification information. The staff then creates a member account in the system. After registration, the member receives a unique membership ID and can start borrowing books according to the library rules.

Book Management

When new books arrive, the librarian adds them to the system with details such as title, author, ISBN, publisher, category, and number of copies. The system should allow staff to update book information whenever needed. If a book is no longer available in the library, it can be marked as removed or inactive instead of deleting its history.

Book Issue Process

When a member requests a book, the librarian first checks the member's status and borrowing limits. The system verifies that the book is available. If it is available, the book is issued to the member and a due date is assigned automatically. The system then records the transaction and updates the book status as "Borrowed."

Book Return Process

When the member returns the book, the librarian searches for the borrowing record and completes the return transaction. The system updates the book status to "Available" and calculates any overdue fine if the return date has passed the due date. The fine is recorded and can be paid immediately or later, depending on library policy.

Lost Book Handling

If a member reports a book as lost, the librarian marks the book as "Lost" in the system. The system should calculate the replacement cost or penalty according to library rules. Once the payment is settled, the case is closed and recorded for future reference.

Damaged Book Handling

If a returned book is damaged, the librarian inspects its condition and records the damage details. Depending on the severity, the member may be charged a repair fee or replacement cost. The book can then be marked as "Under Repair," "Damaged," or "Removed" if it is no longer usable.

Throughout all these processes, the system should maintain a complete history of transactions so that administrators can track member activities, book movements, fines, lost books, and damaged books at any time.

Figure 4: Library Member Requirements

Explain how you expect to search, reserve, borrow, renew, and return books, and what services you expect from the system.



Project Discussion Bot - Year2

Ms. Tharushi Silva – Member Services Coordinator

From a member's point of view, I expect the system to be very easy to use.

For **searching books**, members should be able to search by title, author, ISBN, category, or keyword. The system should show whether the book is available, borrowed, or reserved.

For **book reservations**, if a book is currently borrowed by another member, I expect the member to place a reservation online. The system should notify the member when the book becomes available.

For **borrowing books**, members should be able to see their borrowing eligibility and current borrowing limit. When a book is issued, the system should record the transaction and display the due date clearly.

For **renewing books**, members should be able to request an extension before the due date. The system should check whether the book has been reserved by another member before approving the renewal.

For **returning books**, the system should update the book status immediately after the return is processed. If a book is overdue, any applicable fine should be calculated automatically and shown to the member.

The services I expect from the system include:

- Online book catalogue access
- Advanced book search
- Book reservation facility
- Borrowing and return history
- Book renewal requests
- Fine and payment information
- Due date reminders
- Notifications for reserved books
- Member profile management
- Help and support information

These features will make the library more convenient for both members and library staff while improving the overall user experience.

Figure 5: Finance Officer Requirements

Explain how overdue fines should be calculated, recorded, managed, and what financial reports should be available.



Project Discussion Bot - Year2

Mr. Kasun Wijeratne – Finance and Administration Manager

From the finance side, it is very important that all fines are calculated correctly and recorded properly.

For **overdue fines**, I expect the system to automatically calculate the fine based on the number of days a book is returned late. Our library may decide a fixed charge per day, such as LKR 10 or LKR 20 per overdue day. The system should calculate the total amount automatically when the book is returned.

For example, if the fine is LKR 20 per day and a book is returned 5 days late, the system should calculate a fine of LKR 100.

For recording fines, every fine should be linked to:

- Member ID
- Member Name
- Book ID
- Book Title
- Due Date
- Actual Return Date
- Number of Overdue Days
- Fine Amount
- Payment Status

For **fine management**, the system should allow staff to:

- View all outstanding fines
- Record fine payments
- Issue receipts for payments
- View payment history
- Waive or reduce fines with administrator approval
- Track unpaid fines

The system should clearly show whether a fine is:

- Pending
- Partially Paid
- Fully Paid
- Waived

For **lost or damaged books**, the system should also record:

- Replacement cost
- Repair charges
- Compensation amount paid by the member
- Payment status

The financial reports I would like to have are:

- Fine Collection Report
- Outstanding Fines Report
- Fine Payment History Report
- Lost Book Compensation Report
- Damaged Book Charges Report
- Daily Revenue Report
- Monthly Revenue Report
- Yearly Revenue Report
- Waived Fine Report
- Member-wise Fine Report

These reports should be filterable by date range and should be available for printing or downloading. They help us monitor library income, identify members with unpaid fines, and support proper financial audits and management.

Figure 6: Security and System Requirements

What security features, validations, user permissions, and backup facilities should the system have?



Project Discussion Bot - Year2

Mr. Nuwan Perera – Library Director (Library Administrator)

From an administrator's perspective, security is very important because the system will store member records, book information, borrowing history, and financial data.

For **security features**, I expect every user to log in using a username and password. Passwords should be stored securely. The system should automatically log users out after a period of inactivity. All important activities such as adding books, deleting records, updating member information, and waiving fines should be recorded in an audit log so we can track who performed each action.

For **validations**, the system should prevent incorrect or incomplete data from being entered. For example, mandatory fields such as member name, contact number, and book title should not be left blank. ISBN numbers should be validated, email addresses should follow the correct format, and duplicate member or book records should be prevented whenever possible.

For **user permissions**, I expect different access levels:

Administrator

- Full access to all system functions
- Manage users and permissions
- View all reports
- Approve fine waivers
- Manage system settings

Librarian

- Register and manage members
- Add, update, and search books
- Issue and return books
- Manage reservations
- Record fines and payments

Library Member

- Search books
- View availability
- Reserve books
- View borrowing history
- Request renewals
- View fines and notifications

A user should only access the functions required for their role.

For **backup facilities**, the system should automatically create backups of the database on a regular schedule, such as daily. Administrators should also be able to perform manual backups when needed. Backup files should be stored securely, and the system should provide a restoration facility in case of system failure, accidental deletion, or data corruption.

I would also like the system to maintain:

- Transaction history
- User activity logs
- Backup history records
- Failed login attempt records

These features help protect library data, reduce the risk of data loss, and ensure that the system remains reliable and secure for daily operations.

Figure 7: Future System Improvements

What problems do you face with the current manual library system, and what improvements do you expect from the new web-based system?



Project Discussion Bot - Year2

Mr. Nuwan Perera – Library Director (Library Administrator)

Our current manual system creates several challenges for both staff and members.

One major problem is that book records are maintained in registers and paper files. Searching for a book takes time, especially when the library has a large collection. Sometimes records become incomplete, damaged, or misplaced.

We also have difficulties tracking borrowed books. Staff must manually check borrowing cards and registers to find due dates. This increases the chance of human errors and makes it harder to identify overdue books quickly.

Member management is another challenge. Updating member information, checking borrowing history, and maintaining membership records require a lot of manual effort. As a result, customer service becomes slow during busy periods.

Fine management is also difficult. Staff must calculate overdue fines manually, which can lead to mistakes and disputes. Preparing financial reports takes additional time because information is stored in different records.

Generating reports is one of our biggest problems. To prepare monthly or yearly statistics, staff have to collect information from multiple registers and count records manually. This process is time-consuming and not always accurate.

Because records are paper-based, there is always a risk of losing important information due to damage, misplacement, or disasters.

With the new web-based system, I expect significant improvements.

I expect faster book searching and instant access to book availability information. Member registration and record management should become quicker and more organised.

The system should automate borrowing, returning, renewals, reservations, and fine calculations. This will reduce errors and save staff time.

I also expect real-time reports and dashboards that help management monitor library activities, book usage, overdue items, member growth, and revenue collection.

Members should be able to search books, reserve books, check due dates, and view their account information online without visiting the library.

Most importantly, I expect the new system to improve accuracy, reduce paperwork, increase efficiency, provide better security for data, and offer a better experience for both library staff and members.

Overall, the goal is to transform our library operations from a slow manual process into a modern, efficient, and reliable digital service.

Figure 8: Book Ratings and Reviews Requirements

As a Library Member, how can I rate and review a book I borrowed? Can I edit or delete my own review, and how should the Library Administrator handle inappropriate reviews?

 Project Discussion Bot - Year2

Ms. Tharushi Silva – Member Services Coordinator

As a library member, I would expect a simple and user-friendly review system.

After borrowing and returning a book, the member should be able to open the book page and submit a rating, such as 1 to 5 stars, along with a written review. Only members who have actually borrowed the book should be allowed to submit reviews. This helps ensure that reviews are genuine.

Yes, members should be able to **edit their own reviews** if they want to correct information or update their opinion after reading the book again. They should also be able to **delete their own reviews** if they no longer wish to share them.

For each review, the system should store:

- Review ID
- Member ID
- Book ID
- Rating (1-5 stars)
- Review Comment
- Review Date
- Last Updated Date
- Review Status (Pending, Approved, Rejected)

To maintain quality, I would prefer that reviews go through a moderation process.

The Library Administrator should be able to:

- View all submitted reviews
- Approve reviews before publication
- Reject inappropriate reviews
- Remove offensive or misleading reviews
- View the review history
- Suspend members who repeatedly misuse the review feature

Examples of inappropriate reviews include:

- Offensive or abusive language
- Spam or advertising content
- False or misleading statements
- Content unrelated to the book
- Harassment of other members or staff

The system should also provide a **"Report Review"** option so members can report reviews they believe are inappropriate.

On the book page, members should be able to see:

- Average book rating
- Total number of ratings
- Member reviews
- Date of each review

This feature will help members choose books more easily and encourage the library community to share useful feedback about books.

Figure 9: Member Feedback Requirements

As a Library Member, how can I submit feedback about library services, check its status, and receive a response from the Library Administrator?

 Project Discussion Bot - Year2

Ms. Tharushi Silva – Member Services Coordinator

As a library member, I would expect a simple feedback system that allows me to share my comments, complaints, suggestions, or service requests about the library.

To **submit feedback**, I should be able to log in to my account and complete a feedback form. The form may include:

- Feedback Subject
- Feedback Category (Complaint, Suggestion, Appreciation, Service Request)
- Feedback Description
- Submission Date

After submitting the feedback, the system should generate a unique **Feedback Reference Number** so I can track it later.

To check **feedback status**, I should be able to view a "My Feedback" section in my account. The feedback status could be:

- Submitted
- Under Review
- In Progress
- Resolved
- Closed

I would also like to see the date of the latest update and any comments from the library staff.

For **receiving responses**, the Library Administrator should be able to review the feedback and provide an official response through the system. Once a response is added, the member should receive a notification by email or through the system dashboard.

The **Library Administrator** should be able to:

- View all feedback submissions
- Assign priority levels
- Update feedback status
- Respond to members
- Maintain feedback history
- Generate feedback reports

A typical process would be:

1. Member submits feedback.
2. System generates a feedback reference number.
3. Administrator reviews the feedback.
4. Administrator updates the status to "Under Review."
5. Administrator provides a response or takes corrective action.
6. Status changes to "Resolved" or "Closed."
7. Member is notified and can view the response.

This feature helps improve communication between members and library staff and allows the library to continuously improve its services based on member feedback.